Flame Response Mechanisms Due to Velocity Perturbations in a Lean Premixed Gas Turbine Combustor

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The response of turbulent premixed flames to inlet velocity fluctuations is studied experimentally in a lean premixed, swirl-stabilized, gas turbine combustor. Overall chemiluminescence intensity is used as a measure of the fluctuations in the flame's global heat release rate, and hot wire anemometry is used to measure the inlet velocity fluctuations. Tests are conducted over a range of mean inlet velocities, equivalence ratios, and velocity fluctuation frequencies, while the normalized inlet velocity fluctuation (V'/V_{mean}) is fixed at 5% to ensure linear flame response over the employed modulation frequency range. The measurements are used to calculate a flame transfer function relating the velocity fluctuation to the heat release fluctuation as a function of the velocity fluctuation frequency. At low frequency, the gain of the flame transfer function increases with increasing frequency to a peak value greater than 1. As the frequency is further increased, the gain decreases to a minimum value, followed by a second smaller peak. The frequencies at which the gain is minimum and achieves its second peak are found to depend on the convection time scale and the flame's characteristic length scale. Phase-synchronized CH^* chemiluminescence imaging is used to characterize the flame's response to inlet velocity fluctuations. The observed flame response can be explained in terms of the interaction of two flame perturbation mechanisms, one originating at flame-anchoring point and propagating along the flame front and the other from vorticity field generated in the outer shear layer in the annular mixing section. An analysis of the phase-synchronized flame images show that when both perturbations arrive at the flame at the same time (or phase), they constructively interfere, producing the second peak observed in the gain curves. When the perturbations arrive at the flame 180 degrees out-of-phase, they destructively interfere, producing the observed minimum in the gain curve.
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1 Introduction

Modern low-emission gas turbines employ lean premixed combustion to minimize NO_x emissions. This approach, however, is known to increase a combustion system's susceptibility to self-sustained combustion instabilities [1,2]. These combustion instabilities occur as a result of the coupling between system acoustics and unsteady heat release from the flame. The sustained high amplitude pressure fluctuations decrease system efficiency and compromise structural integrity. To facilitate the prediction of combustion instabilities in the combustor design stage, a thorough understanding of the flame's response to periodic flow disturbances is necessary and has been an active subject of both analytical [3–8] and experimental [9–19] studies. Experimental flame response measurements can be used to guide the formulation of and validate the predictions of analytical flame response models. Once known, the flame transfer function (FTF) can be incorporated as a source term in a thermoacoustic network model to predict combustion instabilities [20,21].

In order to quantitatively describe a premixed flame's response to upstream velocity perturbations, a FTF describing the ratio of normalized heat release and inlet velocity fluctuations is defined as

$$\mathrm{FTF}(f,A) = \frac{Q'(f)/\bar{Q}}{V'(f)/\bar{V}}$$

where $\bar{Q}$ is the time-averaged heat release rate, $\bar{V}$ is the mean velocity of the mixture in the mixing section, $Q'(f)$ and $V'(f)$ are their corresponding fluctuation amplitudes at the forcing frequency, f, and A is the magnitude of the normalized velocity fluctuation, $V'(f)/\bar{V}$.

Flame transfer functions for laminar premixed flames have been studied by a number of researchers [4,7,22]. Fleifil et al. [4] analytically described the response of a laminar premixed flame to flow oscillations and concluded that the flame transfer function was primarily governed by the flame Strouhal number, defined as $\omega R/S_L$. Lieuwen [22] concluded that the flame transfer function of premixed flames was strongly dependent upon three parameters: the Strouhal number, the ratio of the flame length to width, and the flame shape (i.e., conical, inverted wedge, etc.). Similar to results from analytic and experimental investigations of laminar flame transfer functions, Kim et al. [19] experimentally determined that the response of turbulent premixed flames is governed by three parameters: the Strouhal number, the flame length, and the flame angle. Therefore, the dynamics of turbulent and laminar flames are qualitatively similar [23].

Analytical and experimental studies show that for a given perturbation frequency, the gain is constant until a certain perturbation amplitude is reached, at which point the gain begins to de-

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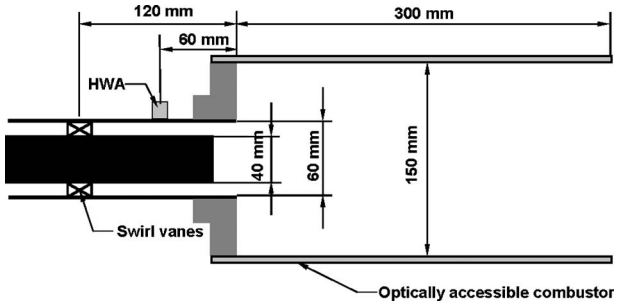

Fig. 1 Schematic diagram of mixing section and combustor

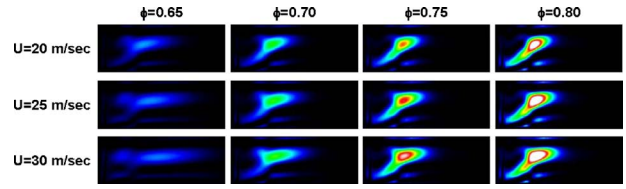

Fig. 2 Deconvoluted unforced flame images (CH^* chemiluminescence) for V_{mean}=20 m/s, 25 m/s, and 30 m/s and ϕ =0.65,0.70,0.75,0.80

crease with increasing forcing amplitude. Correspondingly, the flame response can be divided into a linear regime and a nonlinear regime, respectively [22,24].

In general, the flame behaves as a low pass filter; i.e., the gain decreases with increasing frequency. Schuller et al. [7] analytically examined the flame transfer function behavior for laminar conical and V-flame shapes. While the overall response was that of a low pass filter, the V-flame was also found to exhibit peaks in the gain curve at specific frequencies.

Two flame response mechanisms govern flame dynamics for premixed flames subjected to upstream velocity oscillations: flame perturbations originating from the flame-anchoring point, as reported in Balachandran et al. [11], or perturbations resulting from flow nonuniformities along the flame, described for laminar flames in Durox et al. [10]. Chaudhuri and Cetegen [17] described interactions of these two flame response mechanisms for conical flames. There was, however, no mention of these mechanisms interacting destructively and resulting in a significantly damped global flame response. Preetham and Lieuwen [16] analytically described destructive interactions of two flame perturbation waves, originating from the two flame response mechanisms, which could result in a significantly damped global flame response for both conical flames and wedge-shaped flames.

The objectives of this paper are to experimentally determine the flame transfer function, to identify and quantitatively characterize the mechanisms affecting the flame's response, and to generalize the flame transfer function over a range of inlet velocity and equivalence ratio.

2 Experimental Methods

The experimental measurements are performed in a lean premixed, swirl-stabilized gas turbine combustor facility, utilizing the same combustor and nozzle assembly as Kim et al. [18]. The facility consists of an inlet section, siren modulation device, annular nozzle assembly, and optically accessible quartz combustor section leading to an open exhaust. To eliminate the possibility of equivalence ratio perturbations, air and natural gas are completely premixed before a choked inlet. A siren-based modulation device is used to impose inlet velocity perturbations. Driven by a variable-speed dc motor, the siren is capable of forcing the premixed fuel-air mixture at frequencies ranging from 30 Hz to 500 Hz. Two metering valves control the flow rate of the fuel-air mixture entering the siren device and the siren bypass line. The inlet velocity fluctuation amplitude is varied by changing the fraction of the fuel-air mixture that flows through the bypass line. The annular mixing section consists of a 60 mm inner diameter (i.d.) mixing tube and a 40 mm outer diameter (o.d.) centerbody, as shown in Fig. 1. The centerbody is recessed 20 mm from the dump plane. Within the mixing section, 45 deg swirl vanes are mounted 120 mm upstream of the combustor dump plane. The optically accessible combustor is made of a fully open quartz tube (150 mm in diameter and 300 mm in length).

Measurement instruments include a hot wire anemometer, photomultiplier tubes (PMTs), and an intensified charge coupled device (ICCD) camera. The hot wire anemometer (HWA, TSI model 1210) is used to measure velocity fluctuations in the annular mixing section at 60 mm upstream of the dump plane. Three PMTs (Hamamatsu model H7732-10) are used to measure global chemiluminescence emission intensities from the entire flame. Narrow bandpass filters are mounted on each PMT to allow a separate detection of flame chemiluminescence from CH^* (432 ± 5 nm), CO_2^* (365 ± 5 nm), or OH^* (307 ± 5 nm). Under the premixed conditions of this study, the chemiluminescence intensities provide a measure of the heat release response to inlet velocity fluctuations [25]. To acquire hot wire anemometry and PMT signal data, a digital data acquisition system is used to record 32,768 data points at a sampling rate of 8192 Hz for each test. An ICCD camera (Princeton Instruments model 576 G) equipped with a CH^* bandpass filter (430 ± 5 nm) is used to record flame images. Line-of-sight time-averaged images and phase-synchronized images are taken for a number of experimental conditions. For phase-synchronized images, 40 images are taken per phase angle and 12 phase-averaged images are separated in 30 degree increments of the forcing period with the hot wire signal serving as a reference. To extract a two-dimensional flame structure, a three point Abel deconvolution scheme is applied to the phase- or time-averaged CH^* chemiluminescence line-of-sight images [26].

Experiments are performed at atmospheric pressure with a constant inlet temperature of 23°C. Mean velocities in the nozzle are 20 m/s, 25 m/s, and 30 m/s with mean equivalence ratios (ϕ) ranging from 0.65 to 0.80. The corresponding Reynolds number ranges from 23,700 to 35,600. Normalized flow velocity harmonic forcing amplitudes (V'/V_{mean}) up to 35% can be achieved depending on the forcing frequency. For this study, a fixed forcing amplitude of 5% is used over a frequency range from 80 Hz to 340 Hz.

3 Results and Discussion

3.1 Characterization of Stable Flame Structure. Previous results show that the gain and phase of the flame transfer function depend on the flame structure (flame shape and characteristic flame length) as well as the frequency and magnitude of inlet velocity modulation [6,7,18,19]. Also, flames with the same characteristic flame length exhibit similar flame responses regardless of their inlet flow conditions [19]. The flame length scale is found to be an important parameter determining flame response to inlet velocity perturbations. Time-averaged CH^* chemiluminescence images provide insight into the overall flame structure and the spatial distribution of the flame's heat release. Two-dimensional chemiluminescence flame images are shown in Fig. 2 for a range of mean inlet velocities and equivalence ratios. The flow direction is from left to right, and as mentioned previously, the flame images are deconvoluted, and only the upper halves of the deconvoluted images are shown because the images are axisymmetric. The intensity of chemiluminescence is displayed in pseudocolor, with black being the lowest and white the highest intensity. As the mean velocity and/or equivalence ratio change/s, the flame's shape and length change. Two general flame shapes are observed: typical "V" flames anchored on the centerbody and "V" flames with an extension into the outer corner recirculation zone (CRZ).

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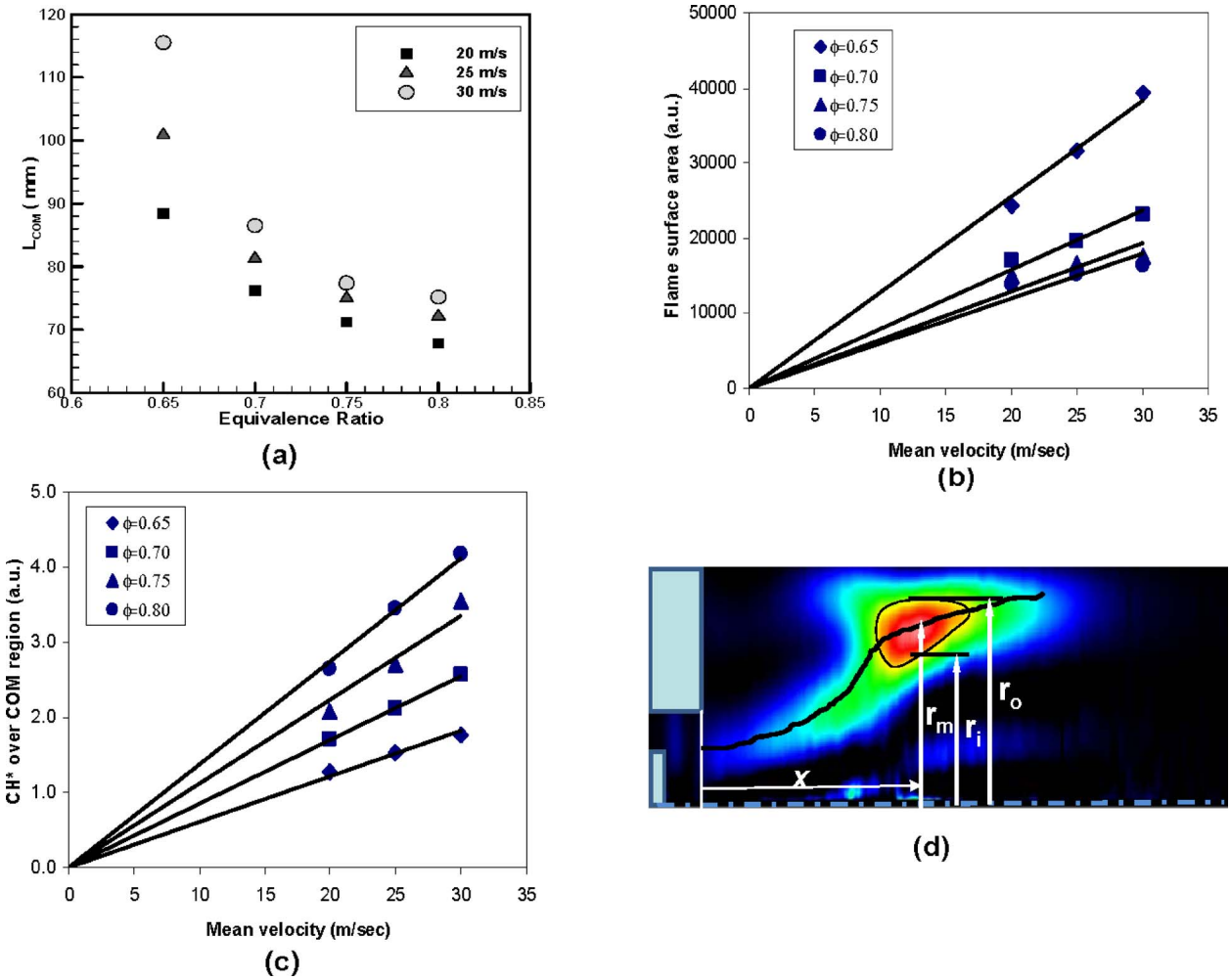

Fig. 3 (*a*) L_{COM} **versus equivalence ratio for various mean velocities,** (*b*) **flame's surface area,** (*c*) **total CH* intensity over COM region versus mean flow velocity for various equivalence ratios, and** (*d*) **a deconvoluted image with the identified mean flame location and the COM region overlaid**

As the flame length is increased, the flame extends into the outer CRZ toward the dump plane, and an earlike structure starts to appear in the CRZ. This is especially apparent at 30 m/s and $\phi = 0.65$. Each image in Fig. 2 shows that the heat release is spatially distributed and that there is a relatively high intensity region at the center of the flame. In a turbulent flame, the flame brush grows with distance from the flame-anchoring location, which results in the broadening of the chemiluminescence flame images along the flow direction.

The high intensity region observed in the chemiluminescence flame images is caused by a significant increase in the flame area due to turbulent wrinkling of the flame surface. This is because for a premixed flame in the flamelet regime the increase in heat release should be accompanied by the increase in flame surface area. Since the overall flame length decreases as the equivalence ratio increases for a fixed mean velocity, the total chemiluminescence intensity over the center part of the flame should increase as the equivalence ratio increases, as can be seen in Fig. 2. As such, the increase in inlet velocity for a fixed equivalence ratio should result in an increase in the flame's length as well as an increase in the total chemiluminescence intensity over the center part of the flame.

In order to quantitatively describe the flame's characteristic length scale, a center of mass (COM) is calculated from the flame image. The distance between the flame COM and the top edge of the centerbody at the dump plane (L_{COM}) can be used to represent the overall length scale or convection length scale of the flame. As shown in Fig. 3(*a*), L_{COM} increases as the inlet mean velocity is increased for the fixed mean equivalence ratio and decreases as the mean equivalence ratio is increased for the fixed inlet mean velocity. Figures 3(*b*) and 3(*c*) show plots of the flame's surface area and total chemiluminescence intensity over the high intensity region versus the mean inlet velocity, respectively. The flame's surface area (S_{flame}) is calculated as follows:

$$S_{\text{flame}} = \int_0^L 2\pi r_m dx$$

from the flame's mean radial location (r_m) identified by locating maximum intensity pixels along the axial distance of each flame image, as shown in Fig. 3(*d*). A region of flame whose intensity is greater than 70% of the maximum intensity is identified using the threshold method for each image, and the sum of pixel intensities weighted by its area over the region is determined as the sum of CH* intensity over the COM region ($Q_{\text{COM region}}$). The flame's surface area linearly increases as the mean inlet velocity increases, and the rate of area increase with respect to inlet velocity increases as the equivalence ratio decreases, suggesting that the heat release per unit surface area is greater for a flame with a higher equivalence ratio. Also, the sum of CH* intensity over the COM region increases as the mean inlet velocity increases, and the rate

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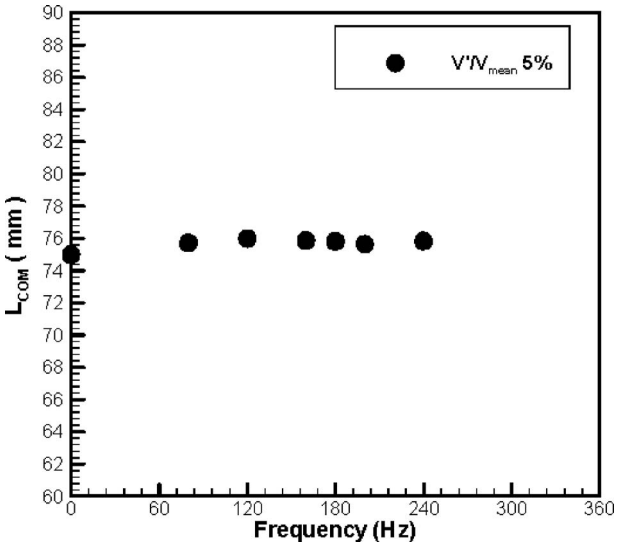

Fig. 4 Frequency dependence of flame length derived from forcing-period-averaged images for V_{mean}=25 m/s and ϕ =0.75

of increase with respect to inlet velocity increases as the equivalence ratio increases. These show that both the flame's surface area and the total chemiluminescence intensity over the high intensity COM region can be used as a measure of heat release. This fact will be used to explain the mechanism of flame response to inlet velocity fluctuation in a later section.

3.2 Characterization of Average Flame Structure Over a Period of Inlet Velocity Fluctuation. The effect of velocity forcing frequency on the averaged characteristic flame length is investigated. An inlet velocity perturbation, which when normalized by the mean velocity (V'/V_{mean}) has an amplitude of approximately 5%, is imposed upon the premixed mixture. This level of harmonic forcing amplitude is chosen because it is small enough to yield a flame response in the linear regime and large enough to provide satisfactory signal coherence between the inlet velocity fluctuation and the chemiluminescence intensity fluctuation. Flame transfer function measurements presented in the next section are determined with the same level of forcing amplitude of 5% of the mean velocity.

As shown in Fig. 4, flame images averaged over each forcing period show that L_{COM} remains relatively constant with respect to forcing frequency. For all inlet flow conditions, unforced flames, which are shown in Fig. 4, at 0 Hz are found to have similar length scales to those of forced conditions. Therefore, the flame transfer function measurements presented in the next section have nearly constant flame length scales with respect to frequency. However, as the flame is subject to greater inlet velocity modulation and responds nonlinearly, L_{COM} is found to decrease with increasing forcing frequency [18].

3.3 Flame Transfer Function Measurement. The flame transfer function is determined for a range of operating conditions with a fixed inlet velocity harmonic forcing amplitude of V'/V_{mean}=5%. Typical flame transfer function results (gain and phase) are shown in Fig. 5 for a mean inlet velocity of 25 m/s and equivalence ratios from 0.65 to 0.8 and in Fig. 6 for an equivalence ratio of 0.7 and mean inlet velocities from 20 m/s to 30 m/s. For a given operating condition, the gain is greater than 1 at low forcing frequencies, indicating that the flame acts as an amplifier at low frequencies. This has been observed in other studies of both laminar [7,16] and turbulent [18,19] V-flames. As the forcing frequency increases, the gain reaches a maximum and then starts to decrease, acting as a low pass filter. As the forcing frequency

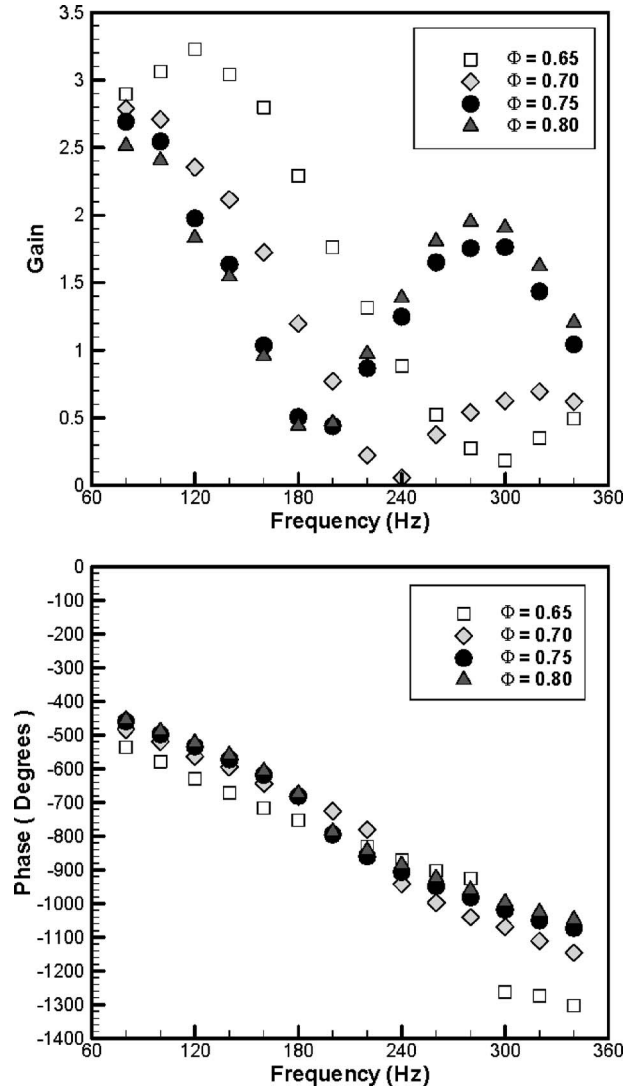

Fig. 5 Fixed forcing V'/V_{mean}=5% FTF gain (top) and phase (bottom) versus frequency for V_{mean}=25 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and forcing frequency=80–340 Hz

increases further, the gain reaches a minimum and then starts to increase again until reaching a second peak. Recalling that the characteristic flame length increases as the equivalence ratio decreases for a given inlet velocity or as the inlet velocity increases for a given equivalence ratio (see Fig. 3), the flame transfer function results indicate that the magnitude of the first peak in the gain plot increases as the characteristic flame length (L_{COM}) increases. A similar behavior has been reported by Kim et al. [19]. The opposite behavior is observed for the second peak where the magnitude of the second peak decreases as L_{COM} increases. Also, the frequency at which the flame response becomes minimum (f_{min}) increases as L_{COM} increases. These results clearly demonstrate that flame length (L_{COM}) affects the flame response to inlet velocity fluctuations.

The lower plots in Figs. 5 and 6 show the phase of the CH^* chemiluminescence intensity fluctuation relative to the inlet velocity fluctuation versus the velocity modulation frequency. The phase increases linearly until the modulation frequency approaches f_{min} at which the phase undergoes a sudden shift of around 180 deg, as predicted by Preetham and Lieuwen [16], after which the phase continues to increase linearly, although at a different rate. The slope of the phase versus frequency curve is directly related to the time between the occurrence of the velocity

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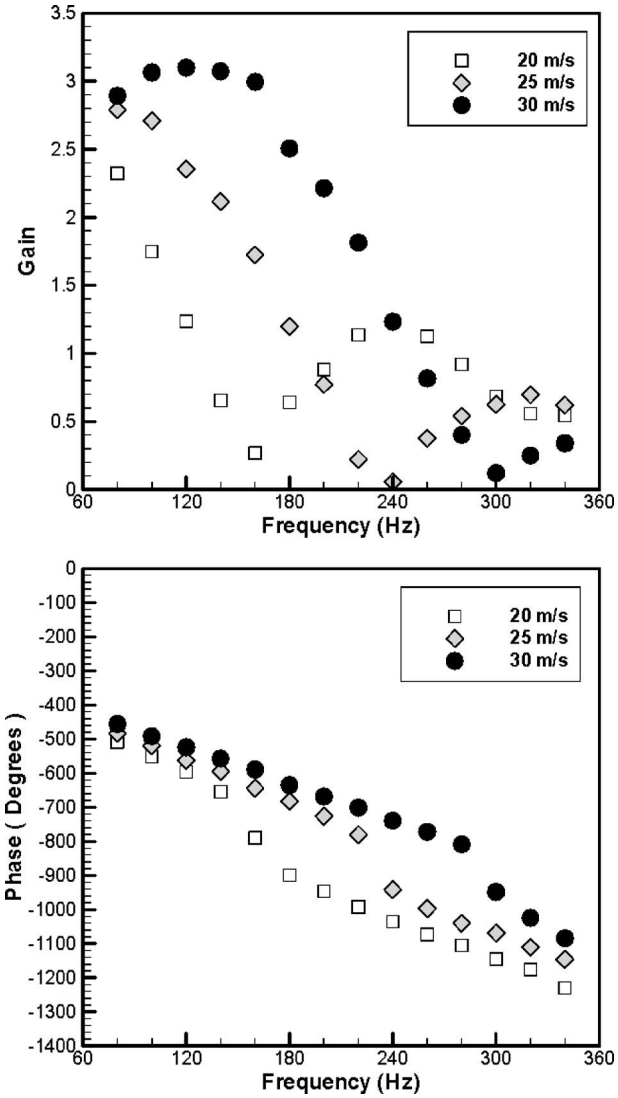

Fig. 6 Fixed forcing V'/V_{mean}=5% FTF gain (top) and phase (bottom) versus frequency for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.70, and forcing frequency=80–340 Hz

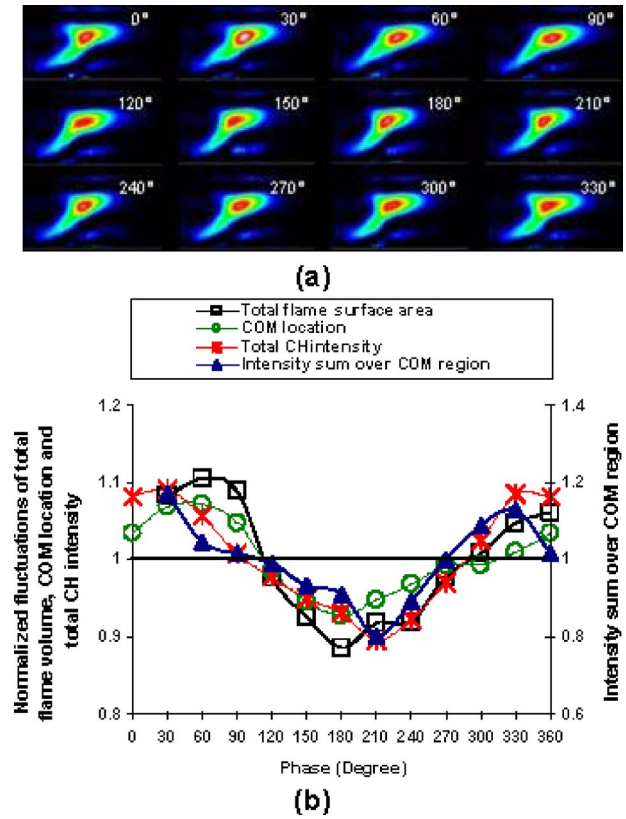

Fig. 7 (*a*) Phase-averaged chemiluminescence images during one period of inlet velocity fluctuation and (*b*) normalized fluctuation of flame area, COM length, total CH* intensity, and sum of CH* intensity over high intensity region (mean inlet velocity=25 m/s, mean equivalence ratio=0.75, and the forcing frequency=100 Hz)

fluctuation (at the two microphone measurement location) and the occurrence of the corresponding fluctuation in the flame's rate of heat release. Thus, there is a marked change in this time when the modulation frequency changes from less than f_{min} to greater than f_{min}.

3.4 Mechanism of Flame Response. Figures 7(*a*) and 8(*a*) show a sequence of phase-averaged flame images recorded in 30 degree phase angle increments at velocity modulation frequencies of 100 Hz and 180 Hz, respectively. In both cases, the mean inlet velocity is 25 m/s, and the equivalence ratio is 0.75. Note that the 100 Hz case corresponds to a condition where the gain of the flame transfer function is greater than 1 (refer to Fig. 5), and the 180 Hz case corresponds to a condition where the gain is at its minimum. In each image, the flow direction is from left to right, and the phase (in degrees) is referenced to the time when the inlet velocity fluctuation is at its minimum. In both cases, it can be seen that the intensity in the central part of the flame and the overall flame length change during the modulation cycle. Figures 7(*b*) and 8(*b*) show plots of the mean flame surface area, center-of-mass location, overall CH* intensity from the whole flame, and the sum of the CH* intensity over the COM region. The flame surface area and the sum of the CH* intensity over the COM region are calculated as described previously. Each quantity is normalized by its mean value. These results show that the global heat release fluctuation (as represented by the CH* intensity) is in phase with the mean flame surface area modulation and the total CH* intensity over the COM region.

Using a kinematic model of flame front evolution, Preetham and Lieuwen [16] studied the dynamics of flames submitted to inlet velocity modulation in terms of the superposition of different perturbations propagating at different velocities: those originating at the flame-anchoring point (such as vortical disturbances) and those due to flow nonuniformities along the flame (acoustic disturbances). The kinematic model showed that the perturbations can either constructively or destructively interact, resulting in an increase or a decrease in the flame's response, respectively. The results presented in this paper provide an opportunity to experimentally study and validate this behavior.

As discussed in the previous section, the overall rate of heat release of a premixed turbulent flame is directly related to the overall flame area, and changes in the overall flame area can be expressed in terms of changes in the mean flame area (or flame length) and changes in the local wrinkling of the flame surface. Similarly, the effect of inlet velocity fluctuations can be represented by two separate phenomena. One is the effect of the velocity fluctuation on the mean flame area or mean flame length, while the second is the effect of the vorticity fluctuations generated at the exit of the injector by the inlet velocity fluctuations on local flame wrinkling [10,11]. Because these perturbations travel from the combustor inlet at different speeds, i.e., the acoustic velocity fluctuation propagates at the speed of sound and the vorticity fluctuation propagates at the mean flow velocity, their combined ef-

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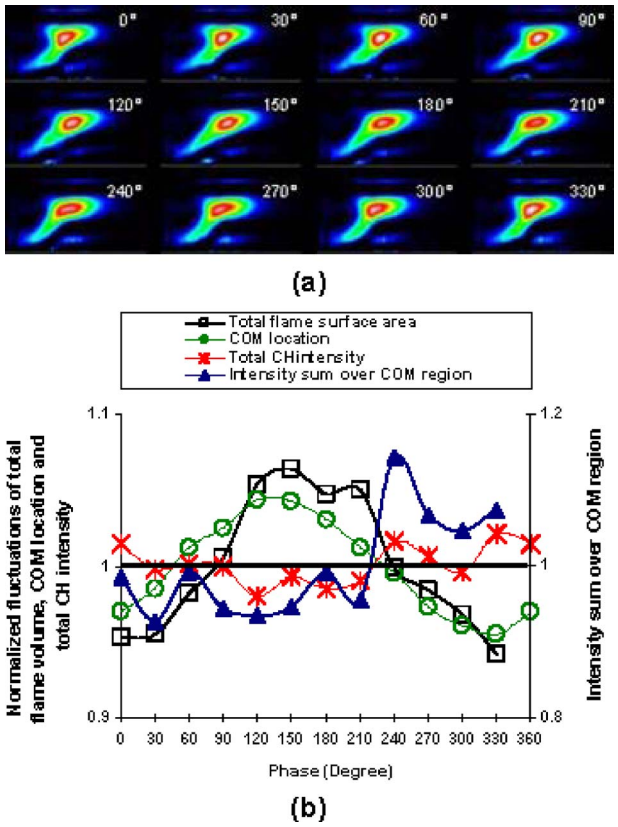

Fig. 8 (*a*) Phase-averaged chemiluminescence images during one period of inlet velocity fluctuation and (*b*) normalized fluctuation of flame area, COM length, total CH* intensity, and sum of CH* intensity over high intensity region (mean inlet velocity=25 m/s, mean equivalence ratio=0.75, and the forcing frequency=180 Hz)

fect on the flame depends on their relative phase when they arrive at the flame. For example, in-phase arrival at the flame will result in constructive interference and increased gain, while if they arrive in out-of-phase they will destructively interfere and result in a diminished gain. The cases of constructive interference and destructive interference are illustrated in Figs. 7(*b*) and 8(*b*), respectively. In Fig. 7(*b*), it is seen that when the flame response is amplified, the heat release fluctuations caused by the two disturbances (the fluctuation in the mean flame area, □, and the fluctuation on the local wrinkling of the flame surface, ▲) are almost in phase, and they result in a greater (10% of mean) amount of overall heat release than the inlet velocity fluctuation (5% of mean). However, Fig. 8(*b*) shows that when the overall flame response is minimal, the heat release fluctuations caused by the two disturbances are almost out-of-phase, indicating that the effect of one on the overall heat release cancels that of the other, and this destructive interaction leads to relatively low overall heat release fluctuation. It should be noted, however, that each perturbation causes a rather significant amount (about 10% of mean) of heat release fluctuation though the overall heat release fluctuation becomes minimal.

In both forcing cases, the overall heat release is associated with the change in L_{COM}. However, at higher forcing frequency cases (f=260 Hz), L_{COM} is found to change minimally while the overall heat release fluctuates significantly (about 8%), as shown in Fig. 9. Also, in that case the heat release fluctuation caused by the two disturbances are in phase, and the overall flame response is amplified but not quite as greatly as that for f=100 Hz.

3.5 Generalization of Flame Response. Figure 10 shows a

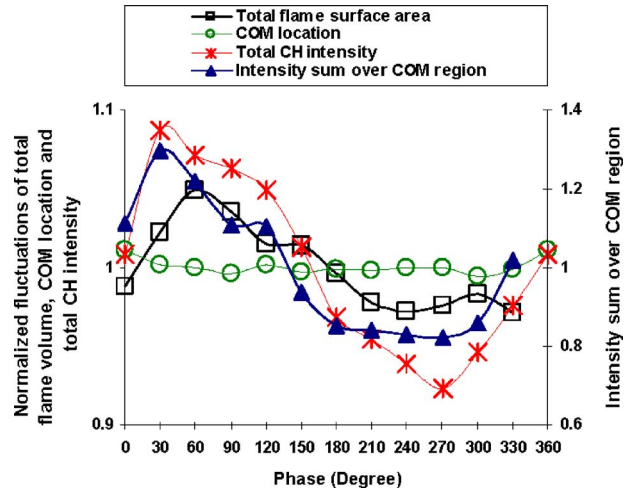

Fig. 9 Normalized fluctuation of flame area, COM length, total CH* intensity, and sum of CH* intensity over high intensity region (mean inlet velocity=25 m/s, mean equivalence ratio =0.75, and the forcing frequency=260 Hz)

plot of f_{min} versus L_{COM}. As observed previously, f_{min} increases as L_{COM} increases. However, the relationship between f_{min} and L_{COM} is distinct for each mean convection velocity: For a given L_{COM}, the higher the mean velocity, the higher the f_{min}. This suggests that the time scale ($\tau_{dip} \sim 1/f_{min}$) associated with the mechanism causing minimum gain at f_{min} is related not only to the characteristic flame length scale but also to the convection velocity. Therefore, a Strouhal number is defined such that τ_{dip} is related to the convection time scale ($\tau_{conv}=L_{COM}/U$) as

$$St_{dip} = \frac{\tau_{conv}}{\tau_{dip}} = \frac{f_{min}L_{COM}}{U}$$

Figure 11 is a plot of St_{dip} versus L_{COM}, showing that τ_{dip} (or f_{min}) can be determined if the characteristic length scale (L_{COM}) and mean flow velocity (U) are known, and St_{dip} exponentially increases with respect to L_{COM}. The latter means that τ_{dip} decreases as L_{COM} increases for flames with the same τ_{conv} and

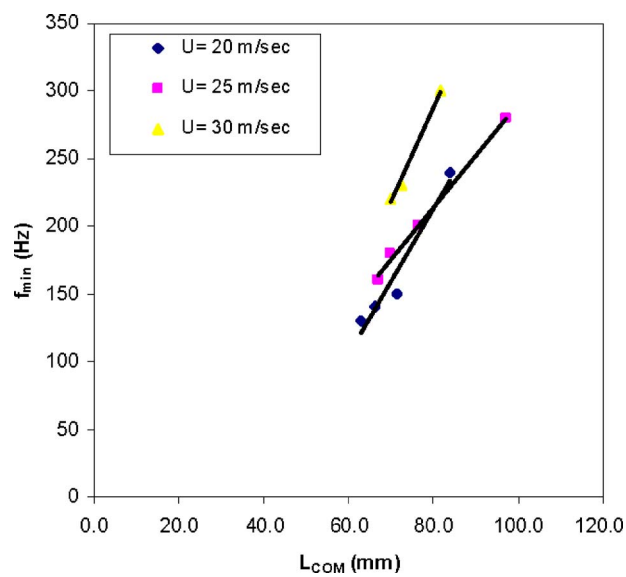

Fig. 10 f_{min} versus characteristic flame length (LCOM) for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

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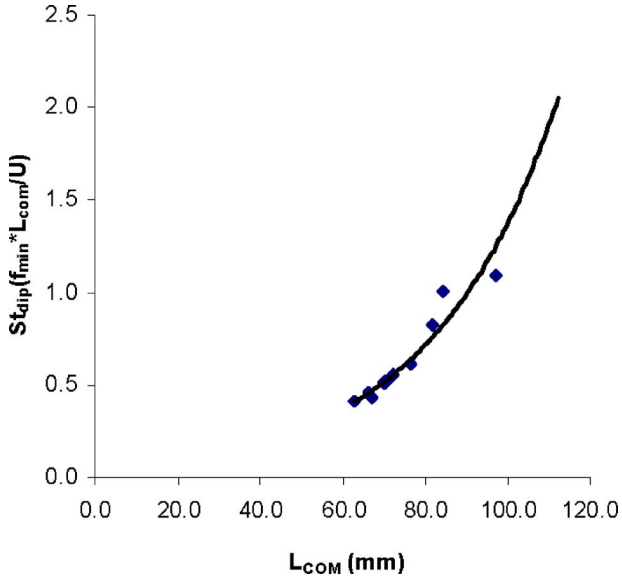

Fig. 11 St_{dip} versus characteristics flame length (LCOM) for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

increases as τ_{conv} increases for flames with the same L_{COM}. As discussed in the previous section, if the minimum gain condition is achieved when the flame area change due to acoustic velocity fluctuations and the flame area change due to vorticity fluctuations are out-of-phase and destructively interfere, then the relationship between St_{dip} and L_{COM} suggests that the velocity scale associated with the former perturbation (i.e., L_{COM}/τ_{dip}) is scaled with the convection time scale and exponentially increases with respect to the characteristic flame length scale.

Previously, when discussing Figs. 5 and 6, it is observed that if the forcing frequency is increased beyond f_{min}, the gain reaches the second peak. Figure 12 is a plot of the frequency at which the second peak occurs versus f_{min}: The higher the f_{min}, the higher the frequency at which the second peak occurs. For f_{min} being correlated with τ_{conv} and L_{COM}, it indicates that the frequency at which the second peak occurs also depends on τ_{conv} and L_{COM}.

Both analytical and experimental flame transfer function studies

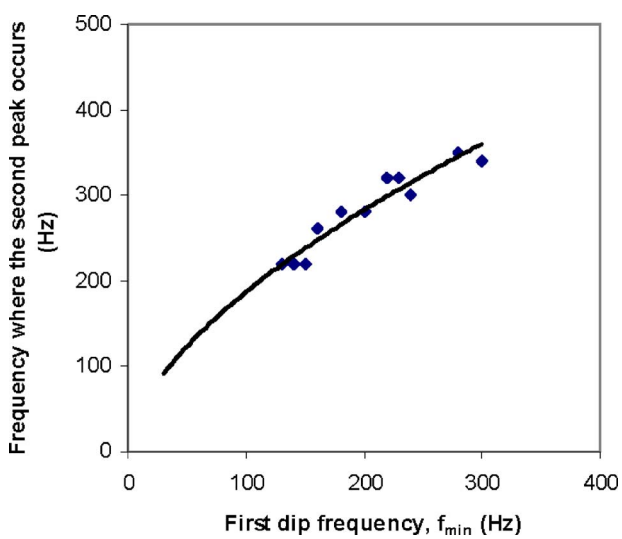

Fig. 12 The frequency at which the second peak occurs versus f_{min} for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

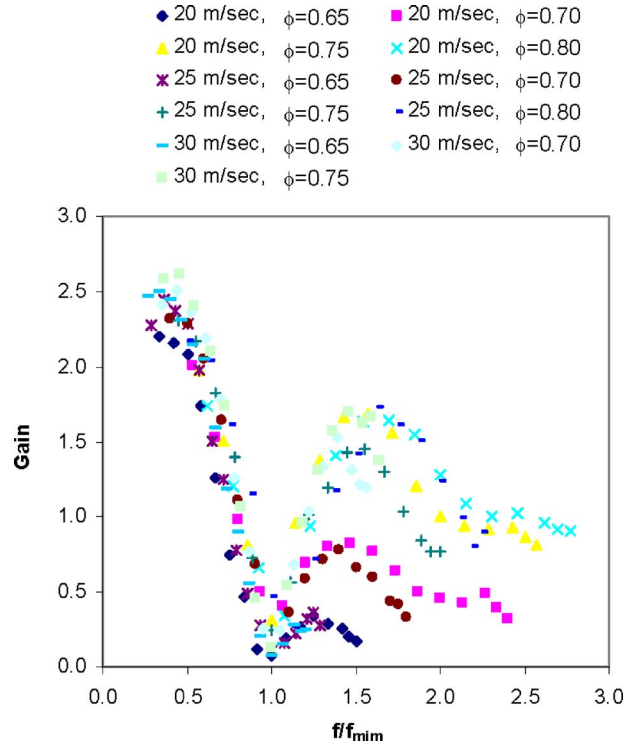

Fig. 13 Gain of flame transfer function versus normalized frequency (f/f_{min}) for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

[3–19] have shown that the gain and phase of the flame transfer function can be generalized in terms of a Strouhal number defined as

$$\mathrm{St} = \frac{L}{L_{conv}} = \frac{L}{V_{eff}/f}$$

where L is a characteristic length scale of the flame or burner, V_{eff} is an effective convection velocity, and f is the forcing frequency. This means that even if the operating conditions are different, the flame response can be determined by the ratio of the two length scales. However, the detailed relationship between the flame transfer function and the Strouhal number is also known to depend on other factors, especially flame shape (i.e., V-flame or conical flames), as discussed in Sec. 1. Figure 13 shows flame transfer function gain measurements for flames where the mean flow velocity is varied from 20 m/s to 30 m/s (20 m/s, 25 m/s, and 30 m/s) and where the equivalence ratio is varied from 0.65 to 0.8 (ϕ=0.65,0.70,0.75,0.80). The frequency is normalized by the frequency at which the gain reaches the first minimum, f_{min}. These results show that the gain can be generalized in terms of the normalized frequency when the perturbation frequency is less than f_{min}, suggesting that the gain of flame response over the frequency range ($f/f_{min}<1$) is governed by the interaction between perturbations originating from the flame-anchoring point and from flow-forcing nonuniformities along the flame, as discussed in the previous section.

When the forcing frequency is greater than f_{min} (or $f/f_{min}>1$), Fig. 13 shows that the magnitude of the second peak varies by about a factor of 6. However, it shows that the magnitude of the second peak has a clear dependency on the characteristic flame length scale, as shown in Fig. 14: In general, the longer the L_{COM}, the smaller the magnitude of the second gain peak, but it is bounded by two limit values. A closer analysis shows that the high limit value is for typical V-flames anchored on the centerbody, and the low limit value is for V-flames with an extension into the outer

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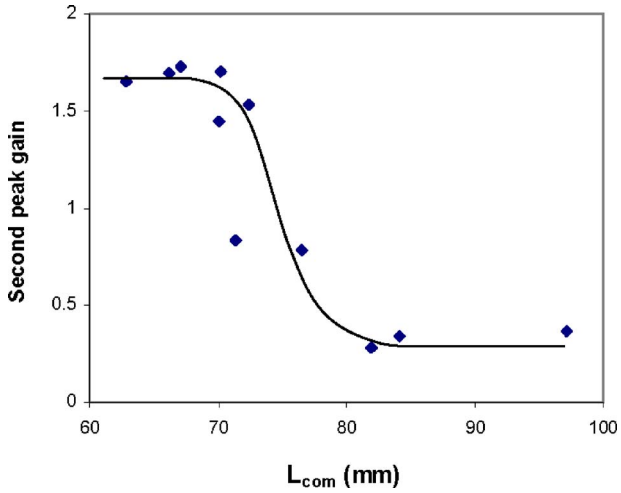

Fig. 14 Magnitude of second peak gain versus LCOM for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

corner recirculation zone.

Figure 15(*a*) shows plots of convection time (τ_{conv}) calculated from the phase plots (such as the ones shown in Figs. 5 and 6) versus L_{COM}. For each phase plot, the data for forcing frequency, which is less than f_{min}, are used to calculate the convection time.

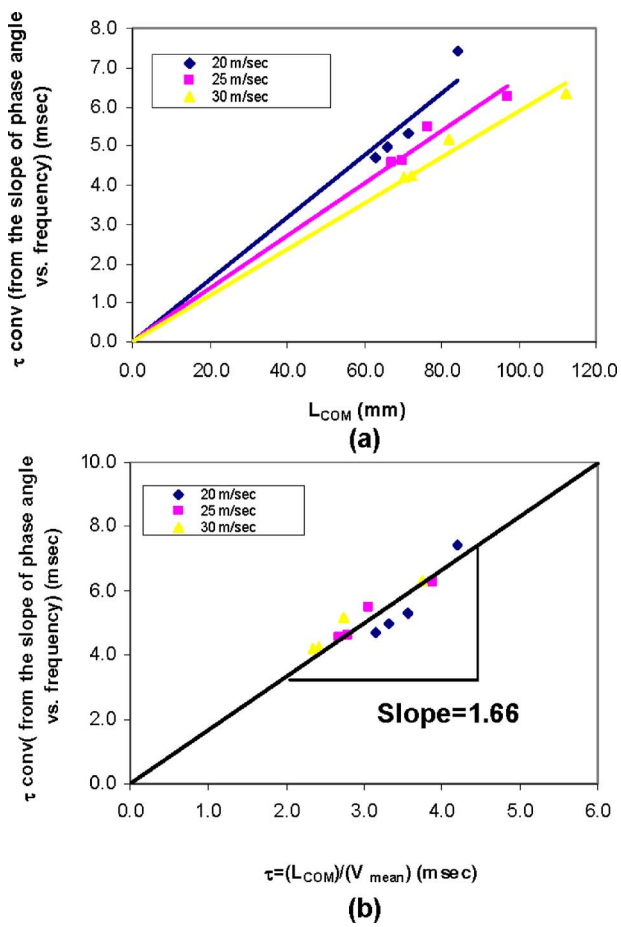

Fig. 15 (*a*) τ_{conv} versus L_{COM} and (*b*) τ_{conv} versus τ (convection time based on L_{COM} and V_{mean}) for V_{mean}=20 m/s, 25 m/s, and 30 m/s, ϕ=0.65, 0.70, 0.75, and 0.80, and V'/V_{mean}=5%

It increases as L_{COM} increases. However, the relationship between τ_{conv} and L_{COM} is distinct for each mean convection velocity: For a given L_{COM}, the higher the mean velocity, the shorter the τ_{conv}. However, Fig. 15(*b*) shows that τ_{conv} is correlated better with convection time calculated using L_{COM} and V_{mean} as the characteristic length and velocity scale, respectively. Since the slope of Fig. 15(*b*) represents the ratio of the mean velocity to the actual convection velocity, it shows that the actual convection velocity is 60% (or 1/slope) of the mean inlet velocity for all operating conditions considered in this study if τ_{conv} is defined in terms of the actual convection velocity and L_{COM}. In other words, the phase can be determined once L_{COM} is known and using 60% of the mean inlet velocity as the characteristic velocity scale.

4 Conclusion

An experimental study is conducted to understand the mechanisms responsible for the linear response of a premixed turbulent flame to inlet velocity perturbations.

The gain of the flame transfer function exhibits the overall low pass filter behavior. However, the gain is amplified at low frequency and becomes a minimum and increases to the second peak as the perturbation frequency increases further. The frequencies at which the gain becomes minimal (f_{min}) and the second peak occurs depend on the convection time scale as well as on the flame's characteristic length scale: The former increases with respect to the flame's characteristic length scale, and the latter increases with respect to f_{min}.

The mechanism resulting in the measured flame response is explained in terms of the interaction between two perturbations one originating at the flame-anchoring point and propagating along the flame front (mean flame surface area) and the other from vorticity field generated in the outer shear layer in the annular mixing section. A detailed image analysis shows that when the flame response is amplified the heat release fluctuation caused by the two disturbances is almost in phase, and they result in a larger (10% of mean) amount of overall heat release compared to that of inlet velocity fluctuation (5% of mean). However, when the flame response is minimal, the heat release fluctuation caused by the two disturbances is almost out-of-phase, and this destructive interaction leads to relatively low overall heat release fluctuation even though each perturbation causes a rather significant amount of heat release fluctuation. This observation indicates that the local flame responses can interact with each other, and a detailed nature of the interaction determines the global response.

The gain of flame response function is generalized over the frequency range of $f/f_{min}<1$ in terms of normalized frequency by the frequency at which the gain becomes minimal, suggesting that the gain of flame response is governed by the interaction between the aforementioned perturbations. The phase is also generalized over the same frequency range in terms of the flame's characteristic length scale and mean inlet velocity.

Nomenclature

A = amplitude
f = modulation frequency
f_{min} = frequency at which the flame response becomes minimum
L = flame length scale
L_{COM} = characteristic flame length (the distance between the flame COM location and the top edge of the centerbody at the dump plane)
Q = heat release from flame
r_m = the flame's mean radial location
S_{flame} = flame's surface area
S_L = lamina flame speed
St = Strouhal number
U = mean mixture velocity
V = inlet mixture velocity

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V_{eff} = effective convection velocity

Greek Symbols

ϕ = equivalence ratio
$\Delta\varphi$ = phase difference between Φ' and $CH^{*\prime}$
ω = angular modulation frequency
τ = time scale

Superscripts and Subscripts

$'$ = amplitude of fluctuation
o = outer dimension
a = inner dimension

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